

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2017 - 2018 (Even Semester)

Innovative Practices Description

Degree, Semester & Branch: VI Semester B.E. ECE A&B

Course Code & Title: EC6001 & Medical Electronics

Name of the Faculty member: Mr.A.Rameshbabu

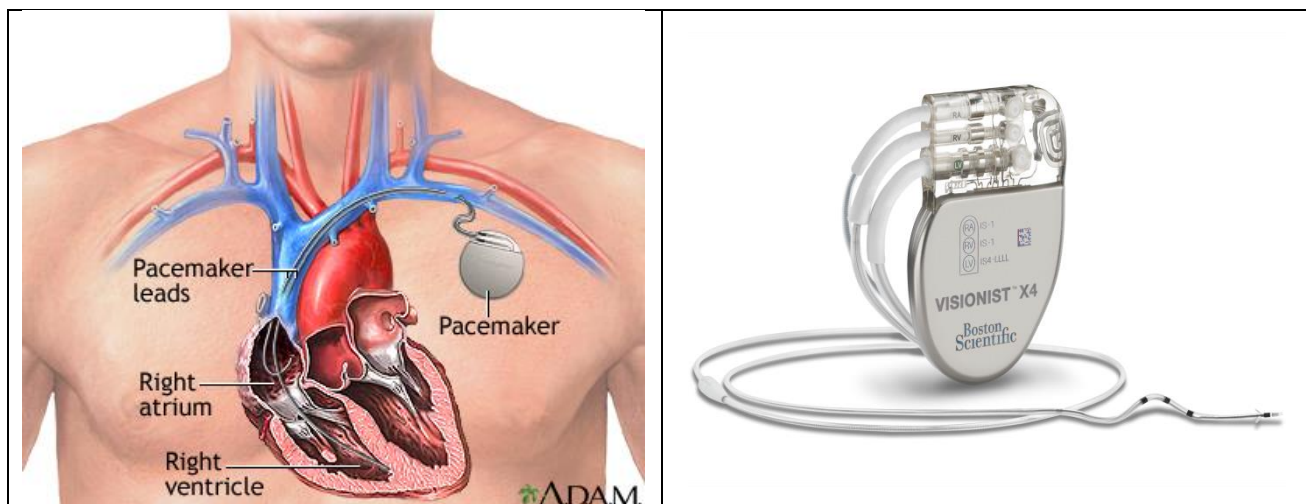
Name of the Topic: Cardiac pacemakers

Name of the Innovative Practice: Animated video

Description:

The contraction of cardiac muscle (heart muscle) in all animals is initiated by electrical impulses known as action potentials. The rate at which these impulses fire, controls the rate of cardiac contraction, that is, the heart rate. The cells that create these rhythmic impulses, setting the pace for blood pumping, are called pacemaker cells, and they directly control the heart rate. They make up the cardiac pacemaker, that is, the natural pacemaker of the heart. In most humans, the concentration of pacemaker cells in the Sino atrial (SA) node is the natural pacemaker, and the resultant rhythm is a sinus rhythm.

Sometimes an ectopic pacemaker sets the pace, if the SA node is damaged or if the electrical conduction system of the heart has problems. Cardiac arrhythmias can cause heart block, in which the contractions lose any useful rhythm. In humans, and occasionally in animals, a mechanical device called an artificial pacemaker (or simply "pacemaker") may be used after damage to the body's intrinsic conduction system to produce these impulses synthetically.



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	1	1	-	-	1	-	-	-	1	1	-	1

CO3: The students will be able to Describe various assist devices used in hospitals.

References:

1. https://en.wikipedia.org/wiki/Cardiac_pacemaker
2. <https://www.youtube.com/watch?v=q8fpQGAD5bk>
3. <https://www.youtube.com/watch?v=22QfCJR8T9M>